Assignment-17.1

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Task 1 – Customer Data Cleaning

Use AI to generate a Python script for cleaning a customer dataset

Prompt:

Perform data preprocessing on the shopping trends dataset by handling missing values, removing duplicates, standardizing text (Location), and encoding categorical variables (Gender, Location).

Code:

assgn17.1 task1.py > ...

```python
#Handle missing values in columns (age, income, city).
#Remove duplicate records.
#Standardize text values (e.g., city names in lowercase).
#Encode categorical variables (gender, city).
import pandas as pd
data = pd.read_csv("C:\\Users\\sushm\\Downloads\\shopping_trends.csv")
# Optional: clean column names (recommended)
data.columns = data.columns.str.lower().str.strip().str.replace(" ", "_")
# Handle missing values
data = data.dropna(subset=['age', 'purchase_amount_(usd)', 'location'])
# Remove duplicates
data = data.drop_duplicates()
# Standardize text
data['location'] = data['location'].str.lower().str.strip()
# Encode categorical variables
data = pd.get_dummies(data, columns=['gender', 'location'], drop_first=True)
print(data.head())
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Myfiles\Attachments\Desktop\Ai-Assist> & C:/Users/sushm/AppData/Local/Programs/Python/Python312/python.e
xe "c:/Myfiles/Attachments/Desktop/Ai-Assist/assgn17.1 task1.py"
   customer_id  age item_purchased  ... location_west virginia  location_wisconsin location_wyoming
0            1   55         Blouse  ...                 False              False           False
1            2   19        Sweater  ...                 False              False           False
2            3   50          Jeans  ...                 False              False           False
3            4   21        Sandals  ...                 False              False           False
4            5   45         Blouse  ...                 False              False           False

[5 rows x 67 columns]
```

Observation:

- Missing values in key columns were removed to improve data quality and avoid errors during analysis.
- Duplicate records were identified and deleted, ensuring no repeated entries affect results.
- The **Location** column was standardized to lowercase and trimmed, maintaining consistency in text data.
- Categorical variables like **Gender** and **Location** were converted into numerical format using one-hot encoding.
- After preprocessing, the dataset became clean, consistent, and suitable for machine learning models and further analysis.

Task 2 – Sales Data Preprocessing

Preprocess sales transaction data using AI assistance.

Prompt:

Perform data preprocessing on the sales dataset by converting the date column into datetime format, extracting new features (month, year, day of week), normalizing the sales amount, and detecting and removing outliers.

Code:

assgn17.1 task2.py > ...

```python
#Convert date columns into datetime format.
# Create new features (month, year, day of week).
#Normalize sales amount column.
#Detect and handle outliers in dataset.
import pandas as pd
data = pd.read_csv("C:\\Users\\sushm\\Downloads\\sales_data.csv")
data.columns = data.columns.str.strip().str.lower()
print(data.columns)  # check once
# Convert date column
data['sale_date'] = pd.to_datetime(data['sale_date'])
# Create features
data['sale_month'] = data['sale_date'].dt.month
data['sale_year'] = data['sale_date'].dt.year
data['sale_dayofweek'] = data['sale_date'].dt.dayofweek
# Normalize
data['sales_amount_normalized'] = (
    (data['sales_amount'] - data['sales_amount'].min()) /
    (data['sales_amount'].max() - data['sales_amount'].min())
)
Q1 = data['sales_amount'].quantile(0.25)
Q3 = data['sales_amount'].quantile(0.75)
IQR = Q3 - Q1
lb = Q1 - 1.5 * IQR
ub = Q3 + 1.5 * IQR
data = data[(data['sales_amount'] >= lb) & (data['sales_amount'] <= ub)]
print(data.head())
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Myfiles\Attachments\Desktop\Ai-Assist> & C:/Users/sushm/AppData/Local/Programs/Python/Python312/python.e
xe "c:/Myfiles/Attachments/Desktop/Ai-Assist/assgn17.1 task2.py"
   product_id  sale_date sales_rep region  ...  sale_month  sale_year sale_dayofweek  sales_amount_normalized
0        1052 2023-02-03       Bob  North  ...           2       2023             4                 0.500950
1        1093 2023-04-21       Bob   West  ...           4       2023             4                 0.433202
2        1015 2023-09-21     David  South  ...           9       2023             3                 0.458201
3        1072 2023-08-24       Bob  South  ...           8       2023             3                 0.209105
4        1061 2023-03-24   Charlie   East  ...           3       2023             4                 0.369108
```

Observation:

- The **sale_date** column was successfully converted into datetime format, enabling time-based analysis.
- New features such as **month, year, and day of week** were extracted, enhancing the dataset with useful temporal information.
- The **sales_amount** column was normalized to a range between 0 and 1, making it suitable for machine learning models.
- Outliers in the sales data were detected using the **IQR method** and removed to improve data consistency.
- After preprocessing, the dataset became structured, clean, and more suitable for analysis and predictive modeling.

Task 3 – Healthcare Data Preparation

Clean and preprocess a healthcare dataset using AI scripts.

Prompt:

Perform preprocessing on health dataset including handling missing values, encoding categorical features, scaling numerical features, and splitting into train and test sets.

Code:

assgn17.1 task3.py > ...

```python
#Handle missing values in blood_pressure, cholesterol.
#Encode categorical features
#Scale numerical features (min-max or standard scaling).
#Split dataset into training and testing sets.
import pandas as pd

# Load dataset
data = pd.read_csv(r"C:\Users\sushm\Downloads\dirty_v3_path.csv")

# Clean column names
data.columns = data.columns.str.lower().str.strip().str.replace(" ", "_")

# Handle missing values
data = data.dropna(subset=['blood_pressure', 'cholesterol'])

# Encode categorical features
data = pd.get_dummies(data, columns=[
    'gender', 'smoking', 'alcohol', 'family_history'
], drop_first=True)

# Split dataset (using medical_condition as target)
X = data.drop('medical_condition', axis=1)
y = data['medical_condition']

# Train-test split
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Myfiles\Attachments\Desktop\Ai-Assist> & C:/Users/sushm/AppData/Local/Programs/Python/Python312/python.exe "c:/Myfiles/Attachments/D
PS C:\Myfiles\Attachments\Desktop\Ai-Assist> & C:/Users/sushm/AppData/Local/Programs/Python/Python312/python.exe "c:/Myfiles/Attachments/D
esktop/Ai-Assist/assgn17.1 task3.py"
X_train: (20400, 19)
X_test: (5100, 19)
y_train: (20400,)
y_test: (5100,)
PS C:\Myfiles\Attachments\Desktop\Ai-Assist>
```

```python
# Train-test split
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Scale numerical features
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()
num_cols = ['age', 'blood_pressure', 'cholesterol']

X_train[num_cols] = scaler.fit_transform(X_train[num_cols])
X_test[num_cols] = scaler.transform(X_test[num_cols])

# Output
print("X_train:", X_train.shape)
print("X_test:", X_test.shape)
print("y_train:", y_train.shape)
print("y_test:", y_test.shape)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Myfiles\Attachments\Desktop\Ai-Assist> & C:/Users/sushm/AppData/Local/Programs/Python/Python312/python.exe "c:/Myfiles/Att
PS C:\Myfiles\Attachments\Desktop\Ai-Assist> & C:/Users/sushm/AppData/Local/Programs/Python/Python312/python.exe "c:/Myfiles/Att
esktop/Ai-Assist/assgn17.1 task3.py"
X_train: (20400, 19)
X_test: (5100, 19)
y_train: (20400,)
y_test: (5100,)
PS C:\Myfiles\Attachments\Desktop\Ai-Assist>
```

Observation:

Missing values were removed, categorical features were encoded, numerical features were scaled, and the dataset was split into training and testing sets for model building.

Task 4 – Employee Salary Data Preprocessing

Clean and preprocess an employee salary dataset using AI scripts.

Prompt:

Perform data preprocessing on the employee salary dataset by handling missing values, detecting and removing salary outliers, standardizing gender values, and encoding categorical features.

Code:

assgn17.1 task4.py > ...

```python
#Handle missing values
#Detect and remove salary outliers.
#Standardize department names (lowercase).
#Apply label encoding to job location, department.
import pandas as pd

# Load dataset (make sure it's CSV, not ipynb)
data = pd.read_csv(r"C:\Users\sushm\Downloads\Employee_Salary_Dataset.csv")

# Clean column names
data.columns = data.columns.str.lower().str.strip().str.replace(" ", "_")

# Handle missing values
data = data.dropna(subset=['salary'])

# Remove salary outliers (IQR)
Q1 = data['salary'].quantile(0.25)
Q3 = data['salary'].quantile(0.75)
IQR = Q3 - Q1

lb = Q1 - 1.5 * IQR
ub = Q3 + 1.5 * IQR

data = data[(data['salary'] >= lb) & (data['salary'] <= ub)]

# Standardize gender
data['gender'] = data['gender'].str.lower().str.strip()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Myfiles\Attachments\Desktop\Ai-Assist> & C:/Users/sushm/AppData/Local/Programs/Python/Python312/python.exe "c:/Myfiles/Attachments/
esktop/Ai-Assist/assgn17.1 task4.py"
   id  experience_years  age  gender  salary
0   1                 5   28       0  250000
1   2                 1   21       1   50000
2   3                 3   23       0  170000
3   4                 2   22       1   25000
4   5                 1   17       1   10000
```

```python
Q3 = data['salary'].quantile(0.75)
IQR = Q3 - Q1

lb = Q1 - 1.5 * IQR
ub = Q3 + 1.5 * IQR

data = data[(data['salary'] >= lb) & (data['salary'] <= ub)]

# Standardize gender
data['gender'] = data['gender'].str.lower().str.strip()

# Encode gender
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
data['gender'] = le.fit_transform(data['gender'])

# Output
print(data.head())
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Myfiles\Attachments\Desktop\Ai-Assist> & C:/Users/sushm/AppData/Local/Programs/Python/Python312/python.exe "c:/Myfiles/Attachments/D
esktop/Ai-Assist/assgn17.1 task4.py"
   id  experience_years  age  gender  salary
0   1                 5   28       0  250000
1   2                 1   21       1   50000
2   3                 3   23       0  170000
3   4                 2   22       1   25000
4   5                 1   17       1   10000
```

Observation:

- Missing values in the **salary** column were removed to ensure data quality.
- Outliers in salary were detected using the **IQR method** and eliminated to avoid skewed analysis.
- The **gender** column was standardized to lowercase for consistency.
- Categorical data (gender) was encoded into numerical format using label encoding.
- The dataset is now clean, consistent, and ready for further analysis or machine learning tasks.